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Cover Letter

Title

Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stratification

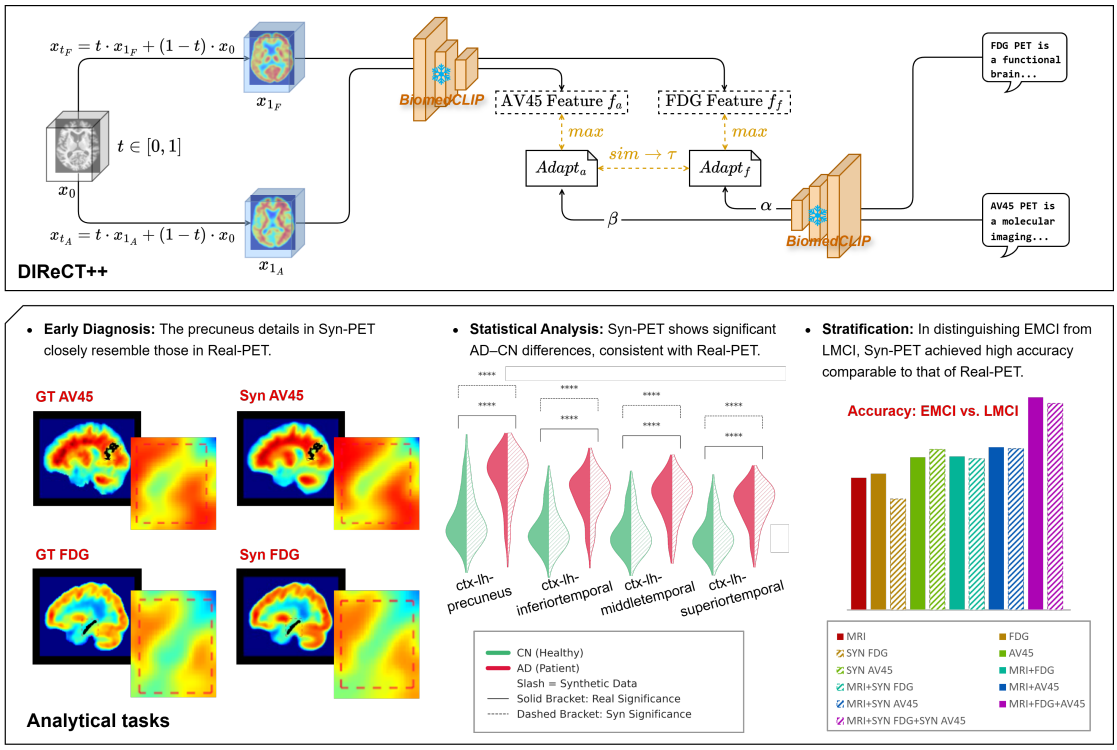
Summary

Dear Editors of *BME Frontiers*,

We are pleased to submit our manuscript entitled “Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stratification” for consideration for publication as a Research Article in *BME Frontiers*. This manuscript is being transferred to *BME Frontiers* following an editorial recommendation from *Research* (Manuscript No. RESEARCH-D-26-03395), where the editors considered the work to be a better fit for the scope of *BME Frontiers*.

Alzheimer’s disease (AD) is a major neurodegenerative disorder in which molecular neuroimaging biomarkers provided by positron emission tomography (PET) play an important role in diagnosis, disease characterization, and clinical research. However, routine application of multi-tracer PET remains limited by acquisition cost, radiation exposure, and accessibility. Developing approaches that can provide PET-like molecular information from widely available imaging modalities represents an important challenge in precision neuroimaging.

In this work, we introduce DIReCT++, a vision-language-modulated rectified flow framework for synthesizing subject-specific multi-tracer PET images directly from a single T1-weighted magnetic resonance imaging (MRI) scan. The proposed framework addresses the inherent ill-posed nature of MRI-to-PET translation by incorporating patient-specific clinical information and tracer-related knowledge through a domain-adapted vision-language model (VLM), enabling knowledge-guided PET generation beyond conventional image-to-image translation.



Using large-scale Alzheimer's disease cohorts, we systematically evaluated the synthesized PET images from multiple perspectives, including image-level fidelity, regional biomarker consistency, and downstream clinical relevance. The generated amyloid and fluorodeoxyglucose (FDG) PET images reproduced disease-associated spatial patterns and preserved information relevant to cognitive impairment assessment. Furthermore, synthetic PET combined with MRI achieved clinical stratification performance comparable to real PET-based approaches, demonstrating the potential of AI-generated molecular imaging for expanding access to PET-related biomarkers.

This study provides a framework that integrates generative artificial intelligence, vision-language modeling, and neuroimaging biomarker analysis. We believe that the proposed approach will be of interest to researchers in artificial intelligence, medical imaging, neuroscience, and precision medicine, particularly those investigating computational methods for noninvasive assessment of neurodegenerative diseases.

Publication Ethics Statement

We confirm that this manuscript has not been published previously and is not under consideration for publication elsewhere, including online platforms. All authors listed in the manuscript have reviewed and approved the submitted version and agree with the publication ethics policies of *BME Frontiers*.

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